

Health monitoring

Application Note

Published by ams-OSRAM AG

Tobelbader Strasse 30,
8141 Premstaetten Austria

Phone +43 3136 500-0

ams-osram.com

© All rights reserved



Health monitoring

Application Note No. AN147

**Valid for:**

BIOFY™ / TopLED™ D5140 / Chip LED™ /
Firefly™ E1608 / Firefly™ E2218 / MULTILED™ / WLCSP

Abstract

This application note explains how special small and compact LEDs, photodiodes and integrated modules enable wearable devices to measure vital signs. Recommended emitters, detectors and integrated solutions are highlighted and key factors for measurement accuracy such as wavelength selection, package design and crosstalk reduction are discussed. Practical system integration tips and performance data from reference designs are also provided.

Table of contents

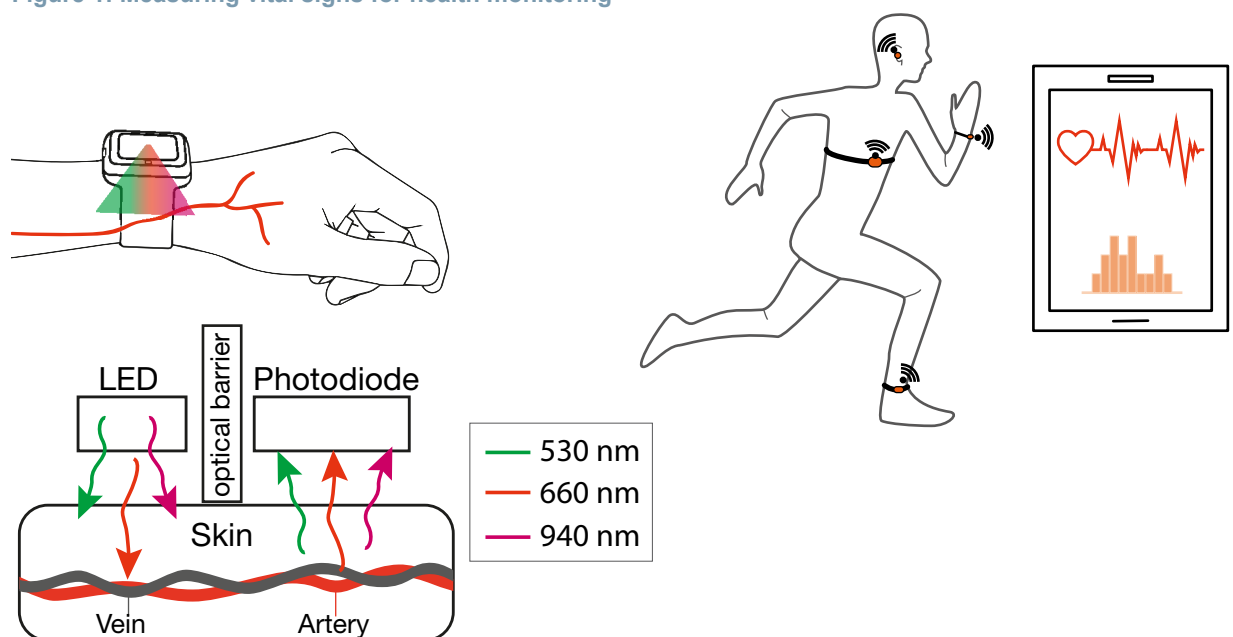
- 1 Basics on monitoring vital signs 4
- 2 Light sources for monitoring systems 5
 - 2.1 Single emitters 5
 - 2.2 Multi emitters..... 6
- 3 Detectors and integrated solutions 9
 - 3.1 Detector recommendation..... 9
 - 3.2 Integrated solutions recommended for health monitoring 13
- 4 Integrated AFE solutions 14
 - 4.1 Analog front-end sensors for vital signs 14
- 5 Practical setup considerations..... 17
 - 5.1 The impact of LED design..... 17
 - 5.2 The impact of light wavelength on measurement result..... 18
 - 5.3 The impact of LED specification 19
 - 5.4 The influence of wavelength shift..... 21
 - 5.5 Operating conditions 23
 - 5.6 Crosstalk reduction 25
- 6 Summary 27
- 7 Glossary and abbreviations..... 28

1 Basics on monitoring vital signs

Wearable applications allow continuous monitoring of vital signs, supporting both medical healthcare and everyday fitness tracking (Figure 1). With each cardiac cycle, the heart pumps blood through the circulatory system. Arteries regularly expand and contract as blood flows through. This caused a volume increase or decrease with each heartbeat. These volume changes can be detected by illuminating the skin with an LED and measuring the amount of light that is either transmitted or reflected using a photodetector. This method allows for the measurement of heart rate, as the periodic changes in blood volume correspond directly to the heartbeat.

This optical principle is also used for measuring blood oxygen saturation. In this case the absorption of the hemoglobin in the blood is measured. Oxygenated hemoglobin (HbO_2) has a significantly different absorption of light than non-oxygenated hemoglobin (Hb). For detection, the skin is illuminated with one red and one IR LED light. Measuring the absorbed light with a photodetector, the Peripheral Oxygen Saturation (SpO_2) value can be calculated as the ratio of the measured signals.

Figure 1: Measuring vital signs for health monitoring



The ams OSRAM product portfolio of LEDs, detectors and ICs provide the technical components for heart rate monitoring, oxygen saturation measurements, and support ECG monitoring with ICs.

2 Light sources for monitoring systems

In wearable health monitoring systems, the optical sensing setup can be realized using either integrated modules or discrete components (single LEDs and photodetectors). Both approaches are equally efficient and capable of delivering precise and accurate photoplethysmography (PPG) measurements. The choice depends on the requirements of the system. Integrated modules combine light emitters and detectors in a single housing, minimizing space requirements. Discrete components enable customized configurations with flexible positioning and combinations, and individually selectable performance parameters.

When choosing the light source, both the LED design and the wavelength of the output light must be considered. Optical barriers between the LED and the photodiode are essential to avoid direct cross talk between emitters and detectors. Cross talk degrades measurement quality. Integrated solutions include barriers, whereas this must be considered when selecting a setup with discrete components.

ams OSRAM provides single color or multi color emitters for using discrete components for heart rate monitoring.

2.1 Single emitters

Due to their point-light emission, single emitters enable precise optical coupling with the skin. Table 1 provides an overview of the recommended emitters. Other LEDs may also be appropriate depending on specific system requirements and design.

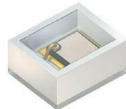
The Firefly™ E1608 provides major benefits for the customer. It enables flexible product designs, due to its compact size. The combination of various wavelengths and typical brightness levels, all matching the application, is possible.

The Firefly™ 2218 provides a more efficient chip in a larger package design for highest system efficiency. This extends the battery life in wearable applications. The CT DBLP31.12 combines high brightness and high efficiency (with biggest advantage at higher currents), whereas the CT DBLP32.12 focuses on maximum brightness.

For further information on these recommended LEDs, please refer to the specific application notes, [“AN088_Firefly™ E1608 and Firefly™ E2218 — Details on handling and processing”](#).

The Chip LED combines a compact footprint with high optical efficiency, making it ideal for space-constrained designs. Its optimized light output supports reliable photoplethysmography (PPG) and pulse oximetry measurements, while the package ensures minimal side emission and stable performance across various operating conditions.

Table 1: Overview of recommended emitters

Emitter	VSM functions and properties	Radiant intensity [mW/sr] typ. @ 20 mA	Package size [mm ³]
Firefly™ 2218	CT DBLP31.12 • HRM	Green: 6.5	2.2 x 1.8 x 0.6 
	CT DBLP32.12 • HRM	Green: 7.5	
Chip LED	CT ELLN51.14 • HRM	Green: 14	1.2 x 1 x 0.6 
	CT DELSS1.12 • HRM	Green: 4	1.6 x 0.8 x 0.6 
Firefly™ E1608	CH DELSS1.22 • SpO ₂	Red: 4.5	
	CN DELSS1.27 • SPO ₂ • PS	IR 940nm: 3.4	
	CN DELSS2.27 • SPO ₂ • PS	IR 850 nm: 3.7	

2.2 Multi emitters

Multi emitters offer the advantage that two or three colors are combined in one package. Table 2 provides an overview on these multi emitters.

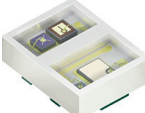
The SFH 7014C, SFH 7014H and SFH 7016 integrate multiple LEDs into a single package. The SFH 7014C is optimized for low-power applications, making it ideal for battery-sensitive designs. In contrast, the SFH 7014H, offers enhanced optical performance and higher brightness.

The SFH 7015 is a compact optical sensor that integrates a red and an infrared LED. The red LED (~660 nm) targets deoxygenated hemoglobin (Hb), while the IR LED (~940 nm) is optimized for oxygenated hemoglobin (HbO₂), enabling accurate SpO₂ calculation.

The MULTILED™ SFH 7018 and is an optimized 3-in-1 solution. The red and IR LED chip are on the same axis and thus have the same distance to the detector. They share a common anode and separated cathodes to drive each LED chip individually. The green LED chip has a separate anode and cathode for an efficient battery management.

The SFH 7019 is a compact 3-in-1 light source combining true green, infrared, and UV-A chips, enabling a wide range of autofluorescence applications. Its chip set also supports healthcare use cases, such as the non-invasive measurement of Advanced Glycation End Products (AGEs) through skin autofluorescence.

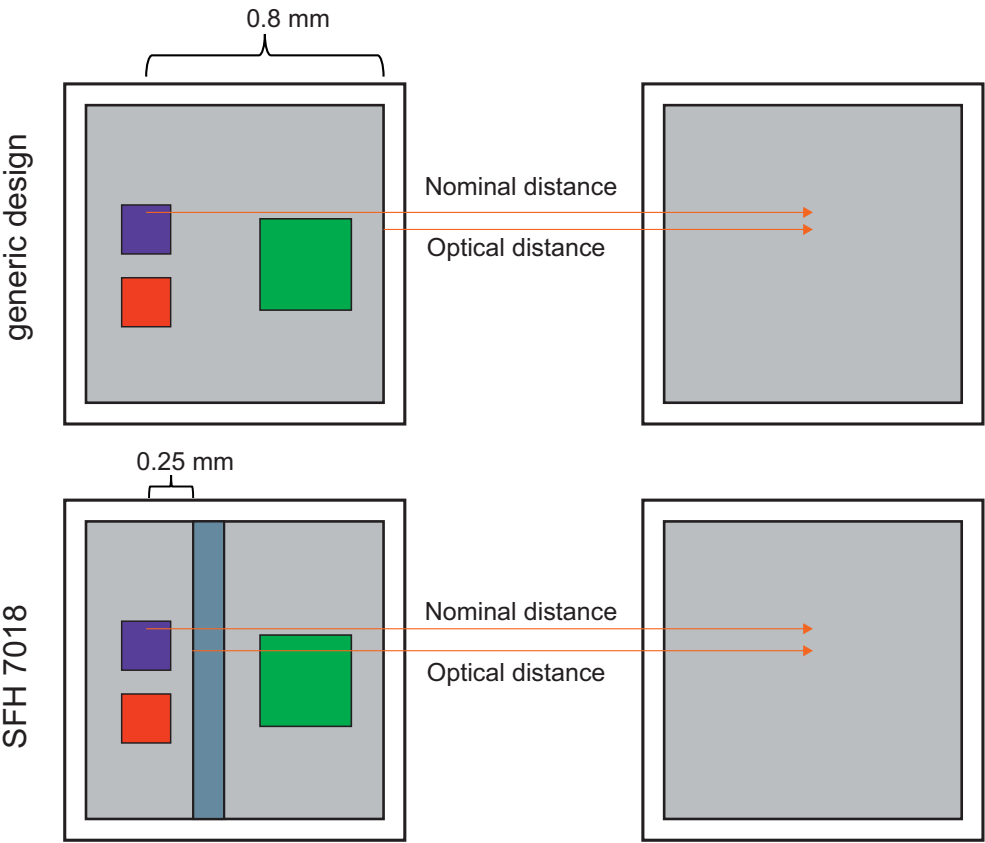
Table 2: Overview of recommended multi emitters

Emitter	VSM functions and properties	Radiant intensity [mW/sr] typ. @ 20 mA	Package size [mm ³]
Multi Chip LED™	SFH 7014C • HRM • SpO₂ • PS	Green: 3.65 Red: 4.2 IR 850nm: 3 IR 940nm: 3	1.85 x 1.65 x 0.6 
	SFH 7014H • HRM • SpO₂ • PS	Blue: 6 Green: 4 Red: 4 IR: 3	1.85 x 1.65 x 0.6 
	SFH 7015 • HRM • SpO₂ • PS	Red: 4.2 IR: 3	2.0 x 0.8 x 0.6 
	SFH 7016 • HRM • SpO₂ • PS	Green: 4 Red: 4.2 IR: 3	1.85 x 1.65 x 0.6 
	SFH 7019 • HRM • SpO₂ • Autofluorescence application (e.g., AGE)	Green: 4.1 UV-A: 5.9 IR: 2.4	2.15 x 1.65 x 0.6 
MULTILED™	SFH 7018 A • HRM • SpO₂ • PS • Light barrier included for reduced crosstalk	Green: 7.7 Red: 6.2 IR: 4.4	2.4 x 2.4 x 0.6 
	SFH 7018 B • HRM • SpO₂ • PS • Light barrier included for reduced crosstalk	Green: 8.5 Red: 6.2 IR: 4.4	2.4 x 2.4 x 0.6 

The SFH 7018 has two cavities, which provide a well-defined optical emission area. Especially for SpO₂ measurements a better definition of the emitter-detector distance is possible as the sketch in Figure 2 explains. In a generic/ standard package design there is a big difference between the nominal distance and the optical distance. The SFH 7018 package offers nearly the same value for the nominal distance as well as the optical distance which optimizes the signal output. It is available in two variants. Variant A offers better V_F for high efficiency, while variant

B has an optimized brightness output. The small package design allows the detectors to be placed at the best possible distance from the light-emitting surfaces.

Figure 2: Light distance



3 Detectors and integrated solutions

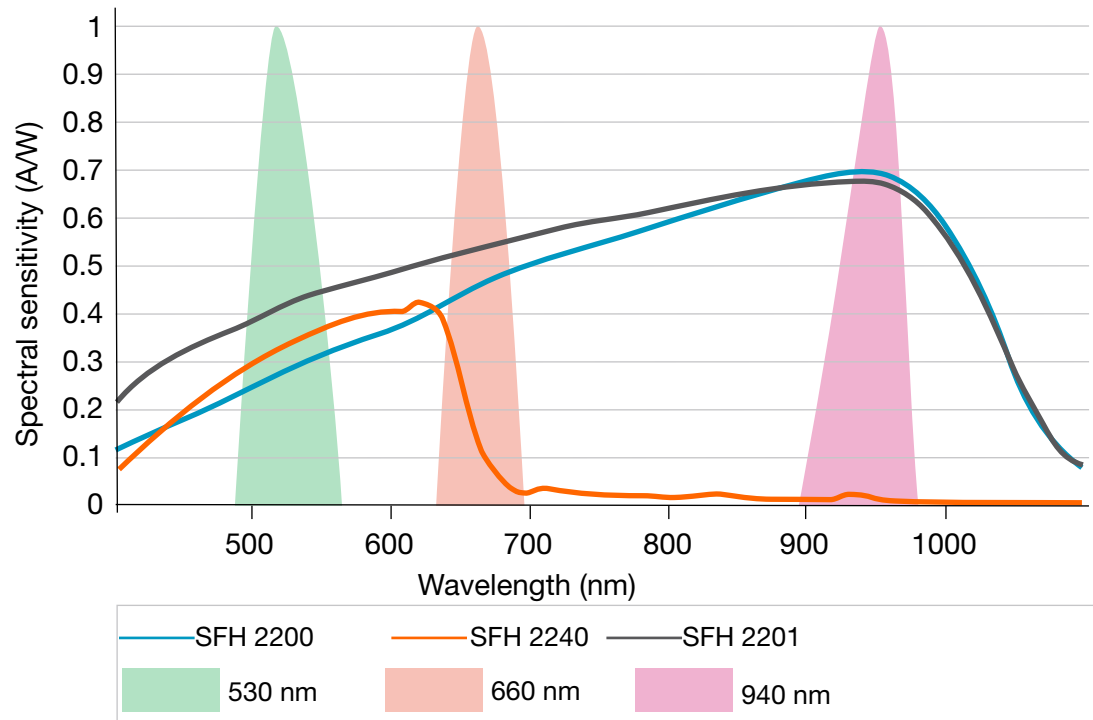
ams OSRAM does not only provides LEDs but also appropriate photodiodes and integrated solutions for health monitoring.

Figure 3 shows the different sensitivity of photodiodes. The SFH 2200 represents a standard photodiode, in comparison to the SFH 2201 (a green enhanced photodiode) and the SFH 2240 (a photodiode with IR-Cut filter).

The green enhanced photodiode SFH 2201 shows a higher sensitivity in green for improved HRM signal detection. Measuring red and IR light is still possible to gain signals for SpO₂ measurement.

Photodiodes with an IR-Cut filter, such as the SFH 2240, are optimized for heart rate monitoring applications. Here the sensitivity to the wavelength range is focused on green. The IR suppression allows noise reduction, which reduces the pulse length and LED current. This leads to an improved HRM measurement and to power savings at the same time.

Table 3: Sensitivity of photodiodes



3.1 Detector recommendation

Detectors for health monitoring systems are available in two packaging formats: TOPLED™ and Chip LED. A key difference is the design of the housing. Detectors of the TOPLED™ family feature a black molded housing, supporting a minimal internal crosstalk. In contrast, Chip LED detectors have a transparent housing. Their compact design allows for a minimal footprint and

flexible integration. However, without optical shielding, attention must be paid to system design to minimize interference.

Recommended photodiodes with a TOPLED™ package (overview Table 4) are the SFH 2200, SFH2201, SFH 2202 and SFH 2240, each offering specific advantages.

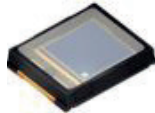
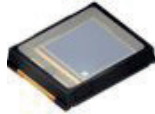
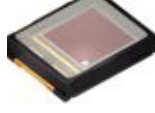
The SFH 2200 is a broadband photodiode without an IR filter, which means it also detects ambient infrared light.

The SFH 2201, provides enhanced green sensitivity. It is well-suited for applications in the spectral range of 300 nm to 1100 nm.

For applications requiring high linearity and broad spectral response, the SFH 2202 is particularly effective, covering wavelengths from 400 nm to 1100 nm. Compared to a standard infrared photodiode, the improved linearity increases the accuracy of SpO₂ measurements and reduces the effects of ambient light variations. The high sensitivity for green allows a reduction of power consumption for heart rate monitoring.

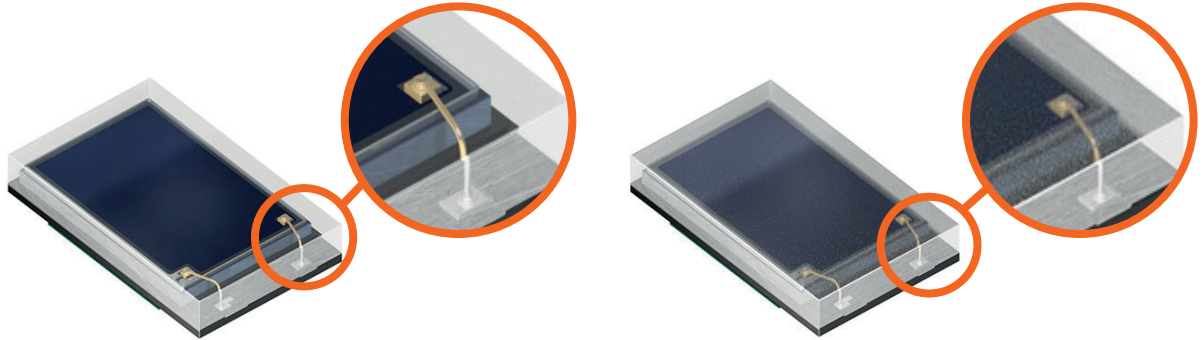
The SFH 2240 is optimized for heart rate monitoring. It features high linearity and fast switching time and suppresses the IR spectrum by an IR cut filter.

Table 4: Overview of recommended detectors (TOPLED™ package)

	Photodiode	VSM functions and properties	Typ. photocurrent I _p [μA] (E _e 0.1mW/cm ² / V _R = 5 V) λ=			Sensitive area [mm ²]	Package size [mm ³]
			530 nm	655 nm	940 nm		
TOPLED™ D5140	SFH 2200	• Broadband • HRM • SpO₂ • PS	2.27	3.21	5.1	7.02	5.1 x 4.0 x 0.85 
	SFH 2201	• HRM • SpO₂ • PS	3.33	3.9	5.03	8.12	5.1 x 4.0 x 0.85 
	SFH 2202	• HRM • SpO₂ • PS • Fast switching time	3.2	4.1	3.8	8.12	5.1 x 4.0 x 0.85 
	SFH 2240	• HRM with IR suppression	2.44	-	-	7.02	5.1 x 4.0 x 0.85 

The Chip LED package offers two types of surfaces - clear or harsh surface- as shown in Figure 3. Clear surfaces are recommended for achieving maximum light output, while harsh surfaces are preferred for applications requiring uniform light distribution.

Figure 3: Package surface (example: SFH 2707 and SFH 2707H)



The SFH 2703 and the miniature broadband photodiode SFH 2704 are broadband photodiodes without an IR filter. Their different package geometries, photodiode sizes and footprints provide more flexibility in application setup. Especially the SFH 2704 is an excellent alternative for applications where board space is a critical parameter. With a height of typ. 0.6 mm it is most suitable for extremely flat assemblies.

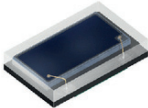
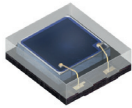
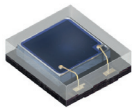
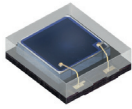
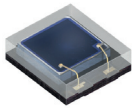
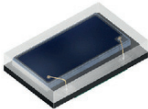
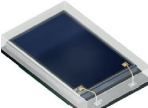
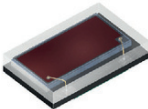
The SFH 2705 and SFH 2706 are photodiodes with an enhanced green sensitivity, especially suitable for applications from 300 nm to 1100 nm. Also here the different package geometries, photodiode sizes and footprints provide more flexibility in application setup. The SFH 2705U offers full footprint compatibility with the SFH 2705 while delivering a significant technological upgrade. Its unique, advanced architecture not only supports standard vital-sign monitoring such as heart rate detection and SpO₂ measurement, but also enables the detection of autofluorescence signals emitted by tissue or materials when excited by a UV light source.

The SFH 2707 is a broadband photodiode designed for a wavelength range of 400 nm to 1100 nm. It offers high sensitivity at 940 nm, very low dark current, and fast response due to its low capacitance.

The SFH 2713 is designed for heart rate monitoring. It features high linearity, fast switching time and suppresses the IR spectrum by an IR cut filter.

Table 5 gives an overview on the recommended photodiodes.

Table 5: Overview of recommended detectors (Chip LED™ package)

Photodiode	VSM functions and properties	Typ. photocurrent I_P [μA] (E_g 0.1mW/cm ² / V_R = 5 V) $\lambda =$			Sensitive area [mm ²]	Package size [mm ³]
		530 nm	655 nm	940 nm		
SFH 2703	• HRM	1.1	1.6	2.3	3.27	3.2 x 2.0 x 0.6
SFH 2703H	• SpO ₂					
	• PS					
SFH 2704	• HRM	0.51	0.77	1.22	1.51	2.0 x 1.8 x 0.6
SFH 2704H	• SpO ₂					
	• PS					
SFH 2704A	• HRM	0.65	0.8	1.2	1.51	2.0 x 1.8 x 0.6
	• SpO ₂					
	• PS					
SFH 2705	• HRM	1.4	1.7	2.3	3.4	2.5 x 2.2 x 0.6
SFH 2705H	• SpO ₂					
	• PS					
SFH 2705U	• HRM	1.35	1.63	2.3	3.4	2.5 x 2.2 x 0.6
	• SpO ₂					
	• PS					
	• Autofluorescence detection					
SFH 2706	• HRM	1.4	1.7	2.3	3.27	3.2 x 2.0 x 0.6
SFH 2706H	• SpO ₂					
	• PS					
SFH 2707	• HRM	1.7	2.1	2.84	4.29	2.0 x 1.8 x 0.6
SFH 2707H	• SpO ₂					
	• PS					
SFH 2713	• HRM with IR suppression	0.95	-	-	3.27	3.2 x 2.0 x 0.6
						

Chip LED

3.2 Integrated solutions recommended for health monitoring

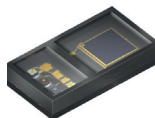
Integrated components provide the Optical Front End (OFE), a combination of a discrete LED and a photodiode, in one device (see also Figure 14).

These integrated components offer the following advantages:

- No external light barrier design needed
- Complex customized optical designs enabled
- Higher light efficacy and potentially lower power consumption
- Fixed emitter-detector distance for good optical consistency (tolerance controlled on the component level)
- Reduced board space
- Simplified SMT process and single component handling

Table 6 provides an overview on the recommended integrated modules.

Table 6: Overview of recommended integrated modules in a multichip package

Module	VSM functions and properties	Radiant intensity [mW/sr] typ. @ 20 mA	Sensitive area [mm ²]	Typ. photocurrent I _p [μA] (E _e 0.1mW/cm ² / V _R = 5 V)				Package size [mm ³]	
				λ =					
				IR Cut PD	530 nm	655 nm	940 nm		
BIOFY™	SFH 7050A	• HRM • SpO₂ • PS	Green: 1.9 Red: 2.9 IR: 2	1.51		0.42	0.76	1.3	4.7 x 2.5 x 0.9 
	SFH 7072	• HRM • SpO₂ • PS	Green: 3.8 (single emitter) Red: 4.8 IR: 3.9	IR-Cut: 3.27 Broadband: 0.88	1.1	0.4	0.6	1.1	7.5 x 3.9 x 0.9 
	SFH 7074	• HRM • SpO₂ • PS • on-chip optical grade IR-cut filter	Green: 4.0 (single emitter) Red: 4.6 IR: 3.0	IR-Cut: 2 x 3.27 Broadband: 3.27	1.9 *	1.1	1.6	2.3	12 x 4.1 x 0.8 

* 0.95 μA each PD

4 Integrated AFE solutions

The integration of sensors into wearables enables overall monitoring of physiological conditions. They enable the creation of a platform for advanced health monitoring in compact form factors.

4.1 Analog front-end sensors for vital signs

Table 7 provides an overview of the AFE components for bio-signal acquisition and highlights their technical specifications.

Table 7: Overview of bio signal converting units

Product	max. current	Headroom @ max. current	Characteristics	Package size [mm³]	
AS7056	200 mA	300 mV	No. of LED drivers: 2 / 2 1x VCSEL Photodiode input: 3 Embedded ECG*: No Proximity: Yes PPG ADC numbers / resolution: 1/19 bit	1.68 x 1.68 x 0.4 	
AS7057 AS7057A	200 mA	300 mV	No. of LED drivers: 3 / 3 Photodiode input: 3 Embedded ECG*: No Proximity: Yes PPG ADC numbers / resolution: 2 /20 bit	1.68 x 1.68 x 0.4 	
WLCSP	AS7058	300 mA	600 mV	No. of LED drivers: 2 / 8 programmable for VCSEL Photodiode input: 8 Embedded ECG*: Yes Proximity: No PPG ADC numbers / resolution: PPG 2 /20bit	2.82 x 2.55 x 0.5 
	AS7058A	300 mA	600 mV	No. of LED drivers: 2 / 8 programmable for VCSEL Photodiode input: 8 Embedded ECG*: Yes Proximity: No PPG ADC numbers / resolution: ECG / GSR/ BioZ 1/ 20bit	2.82 x 2.55 x 0.5 

*: medical grade IEC 60601-2-47 201.12.4.4.108

The AS7058 analog front-end (AFE) represents a state-of-the-art solution for bio-signal acquisition, enabling comprehensive health monitoring through photoplethysmography (PPG), electrocardiography (ECG), galvanic skin response (GSR), and bioimpedance (BioZ). It features two 20-bit PPG ADCs for simultaneous photodiode acquisition, achieving signal-to-noise ratios (SNR) up to 120 dB with post-processing. The device supports eight photodiode inputs and two emitter current sinks, each capable of driving up to 300 mA for LEDs and 25 mA for VCSELs. A 20-bit ECG/BioZ DAC, advanced offset compensation, and ultra-low power consumption (down to 20 μ A at 25 Hz sampling) make the AS7058 ideal for compact, battery-efficient designs.

To validate the AS7058's capabilities, a special wristband (watch V4) was developed (see Figure 4). It integrates the AS7058 AFE, two SFH 7018 multi-chip LEDs, and three SFH 2705 photodiodes enabling the recording of PPG, ECG, GSR, and BioZ signals. It provides a basis for evaluating algorithms.

Figure 4: Wirstband V4



AFE: AS7058
PD: 3x SFH 2705
LED: 2x SFH 7018

Table 8 shows the distances of the center of each LED to the photodiode. The center photodiode was used for green light detection, the further away photodiodes were used for red and IR light.

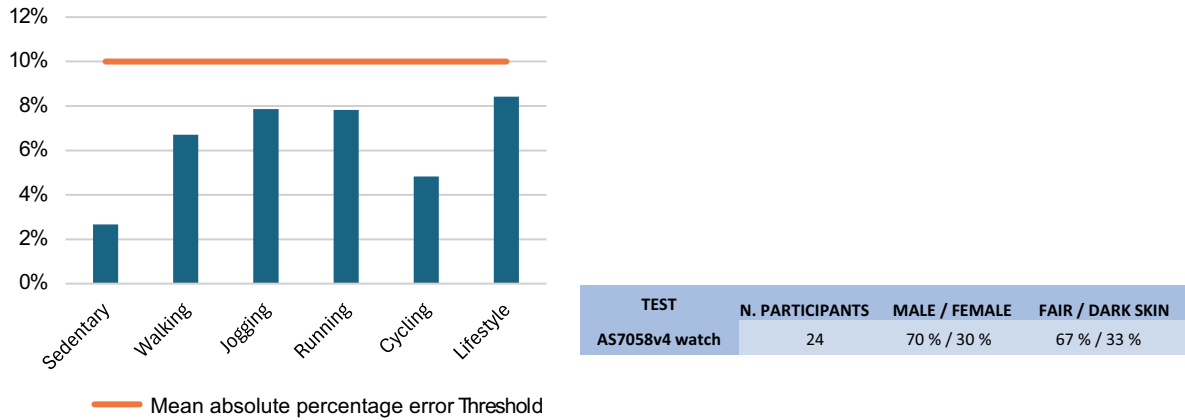
Table 8: Distance LED to photodiode

LED	Distance
Green	3.5 mm
Red	8.5 mm
IR	8.5 mm

For heart rate measurements, the Polar H10 chest-band was used to measure heart rate monitoring accuracy. A study with 24 participants was performed. The demographic composition were 70% male and 30% female and the skin tone distribution of 67% fair and 33% dark skin. Each test session lasted between 6 and 12 minutes per person, depending on the activity. Activities were structured into multiple intensity phases.

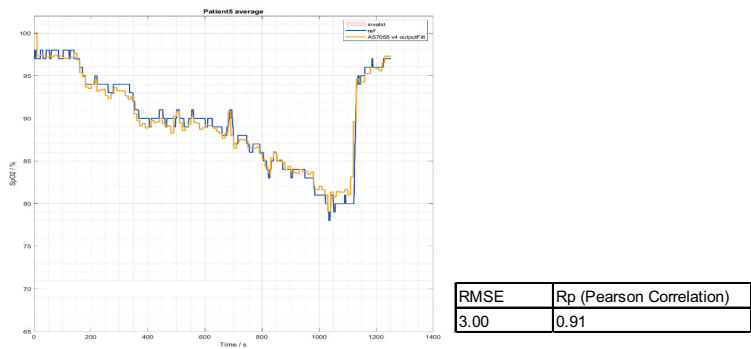
The results (Figure 5) show that heart rate measurement (HRM) is in compliant with ANSI/CTA-2065-A, and the algorithm consistently remains well below the 10% threshold across various activities.

Figure 5: Performance data evaluated with watch V4 for heart rate monitoring



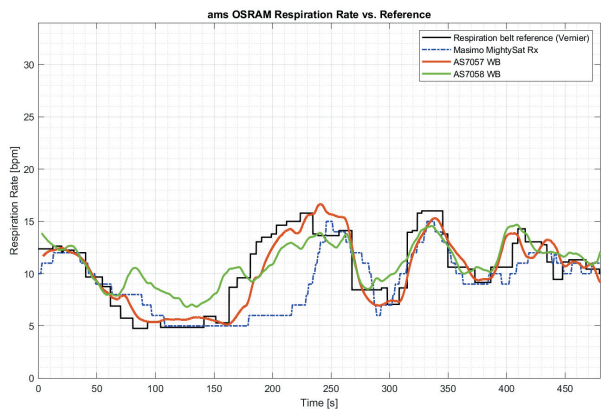
The blood oxygen saturation (SpO₂) performance evaluation was performed by using the watch V4 with 10 participants, of which 40% had dark skin tones (Fitzpatrick scale 4–5). Oxygen levels were regulated through an oxygen mask under medical supervision. The results (Figure 6) show, that the SpO₂ meets FDA requirements for reflective measurements, with RMSE values below 3.5%.

Figure 6: Performance data evaluated with watch V4 for blood oxygen saturation



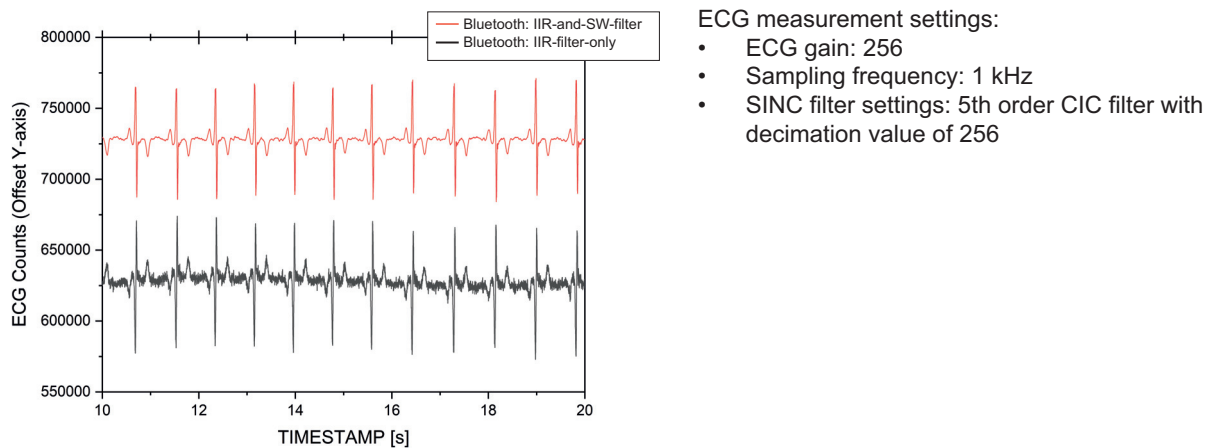
Also the performance of the respiratory rate (RRp) was evaluated using the watch V4 with a vernier respiration belt as reference. Based on the PPG signals, the respiratory rate (RRp) demonstrates performance comparable to that of medical devices (Figure 7).

Figure 7: Performance data evaluated with watch V4 for respiratory rate



Although no algorithm is provided for ECG and BioZ measurements, raw signals are recorded. For ECG measurements in compliance with IEC 60601-2-47 can be achieved depending on the integration of these data. Using the watch V4 setup, tests confirmed that the signals maintain stability and clarity across different wearing positions, ensuring reliable waveform acquisition (Figure 8). For further information on algorithms enable accurate, non-invasive monitoring can be found on [ams-OSRAM website](https://www.ams-osram.com).

Figure 8: ECG data with applied filtering using the watch V4



5 Practical setup considerations

Currently, there are no established standards regarding the skin tolerability of LEDs. Therefore, the application setup must have a transparent cover between the monitoring device and the skin. This cover also protects the assembly against dust and water. However, it may affect the measurement quality due to additional crosstalk.

The signal level (signal quality) is affected by the measurement system as well as by biological characteristics. The use of multiple LEDs can help to achieve reliable measurement results. It is important to maintain a distance of several millimeters between the LED and the photodiode, but the ideal distance for a good signal also depends also on the application system.

The best measurement results can be achieved placing the monitoring system as close to the skin as possible. Additionally, the measurement spot should have good blood circulation. Good circulation results in a noticeable change of the blood volume within the arteries improving the accuracy and stability of the measurements.

5.1 The impact of LED design

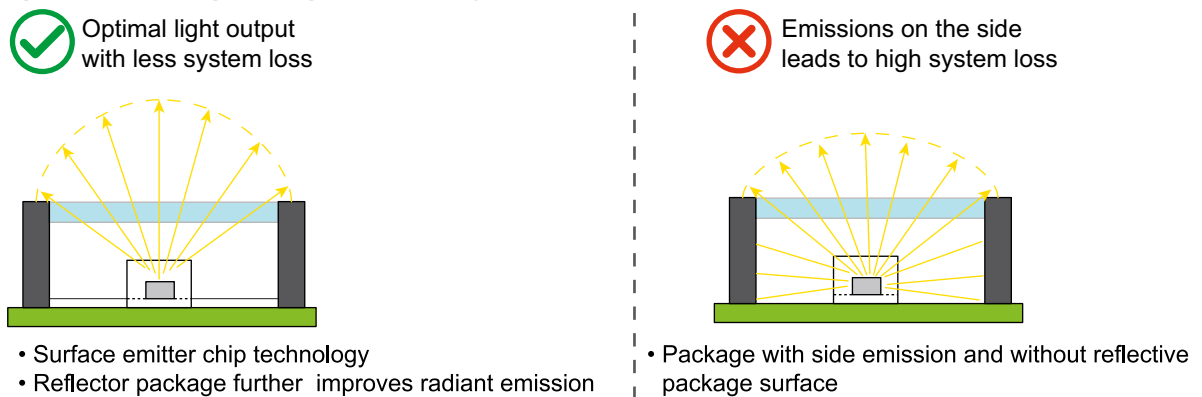
An influencing factor on the efficiency of the LED is the chip size. The larger the chip (at the same current) the smaller the current density. This results in more efficient operation and a lower V_F .

For reliable measurement results, it is also important to use the maximum possible light output. LEDs with top emitting chips reduce the light side emissions and reflective packages further reduce side emissions. LED packages with side emissions will lead to a significant LED light output loss in the application, so surface emitting LEDs with a reflective package are recommended.

For example, the SFH 7018 offers two cavities (integrated barrier) for optimized light output. Integrated components already consider these characteristics.

Figure 9 illustrates the impact of the LED design on the efficiency output.

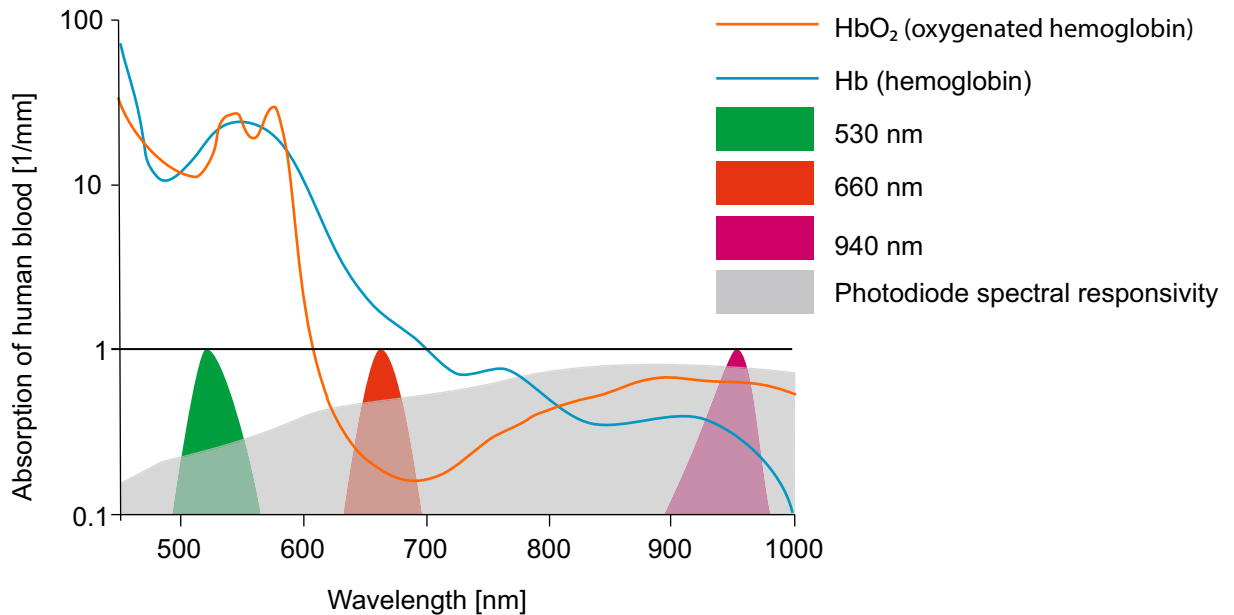
Figure 9: LED design for highest efficiency output



5.2 The impact of light wavelength on measurement result

Absorption of light in human blood mainly depends on hemoglobin. Figure 10 shows the absorption spectra of human blood (hemoglobin (Hb) and oxygenated hemoglobin (HbO₂)) versus the wavelength of light. It also includes the spectral responsivity of the photodiode (PD).

Figure 10: Absorption of human blood versus wavelength of light



Short wavelengths (from blue to yellow) are absorbed strongly. Green is therefore the best choice for heart rate measurement applications, although red and infrared can be successfully used in body locations with a higher concentration of arterial blood (e.g., fingertips, ears, and forehead). Light with longer wavelengths (from red to infrared) is well suited for pulse oximetry measurements. While red light with wavelengths around 660 nm has a higher absorption for non-oxygenated hemoglobin (Hb), infrared light with wavelengths around 960 nm has a higher absorption rate for oxygenated hemoglobin (HbO₂). As a result of the different absorption levels (Hb vs. HbO₂) the SpO₂ level can be calculated by the following formula:

$$SpO_2 = \frac{HbO_2}{Hb + HbO_2}$$

5.3 The impact of LED specification

The following definitions are important in order to understand the impact of the LED specification on the overall device performance in a photoplethysmography application:

- The peak wavelength (λ_{peak}) is defined as the wavelength peak of the spectral density curve (in most applications it is of limited significance).
- The full width at half-maximum (FWHM, $\Delta\lambda$), also referred to as spectral bandwidth, is the wavelength distance between the spectral points where the spectral density $S(\lambda)$ falls to 50 % of the peak value.
- The center wavelength ($\lambda_{0.5m}$) is determined as the wavelength halfway between the two spectral points with spectral density of 50 % of the peak value.

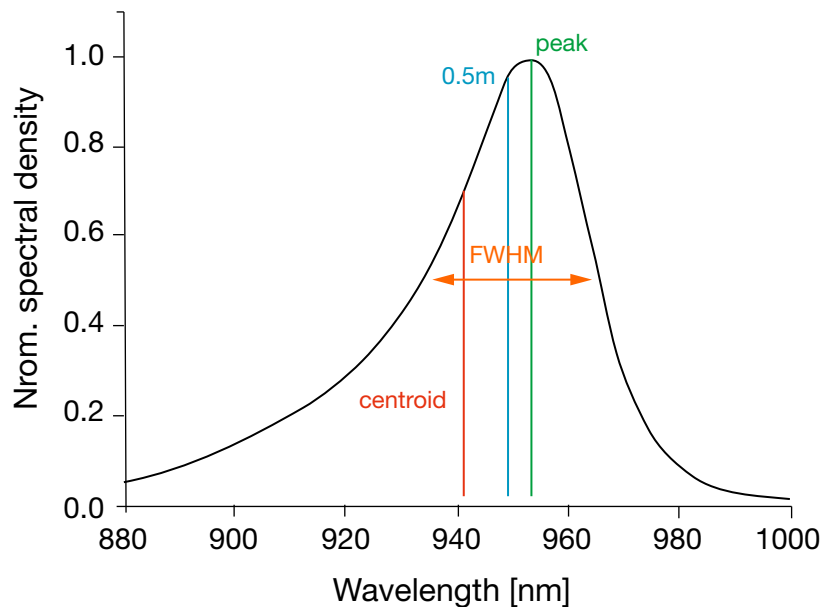
- The centroid wavelength ($\lambda_{\text{centroid}}$) is the mean wavelength. It divides the spectrum into two equal parts. It is the most important definition for non-visual systems (like sensors) and important for photoplethysmography applications. It can be calculated with the following equation:

$$\lambda_{\text{centroid}} \equiv \frac{\int_{\lambda_1}^{\lambda_2} \lambda \cdot S(\lambda) d\lambda}{\int_{\lambda_1}^{\lambda_2} S(\lambda) d\lambda}$$

- The dominant wavelength (λ_{dom}) is a colorimetric quantity and is used to describe the human perception of the color of an LED (within the visible spectrum) for visual illumination systems.

For a symmetrical spectrum λ_{peak} , $\lambda_{0.5m}$ and $\lambda_{\text{centroid}}$ are identical. However, highly efficient LEDs can feature slightly asymmetrical spectra, for example the LEDs inside the SFH 7050. Here λ_{peak} as well as $\lambda_{\text{centroid}}$ are defined, with $\lambda_{\text{centroid}}$ being the important wavelength concerning the application, i.e. the blood absorption coefficients. Figure 11 shows an example of the asymmetric spectrum of the 940 nm IR LED.

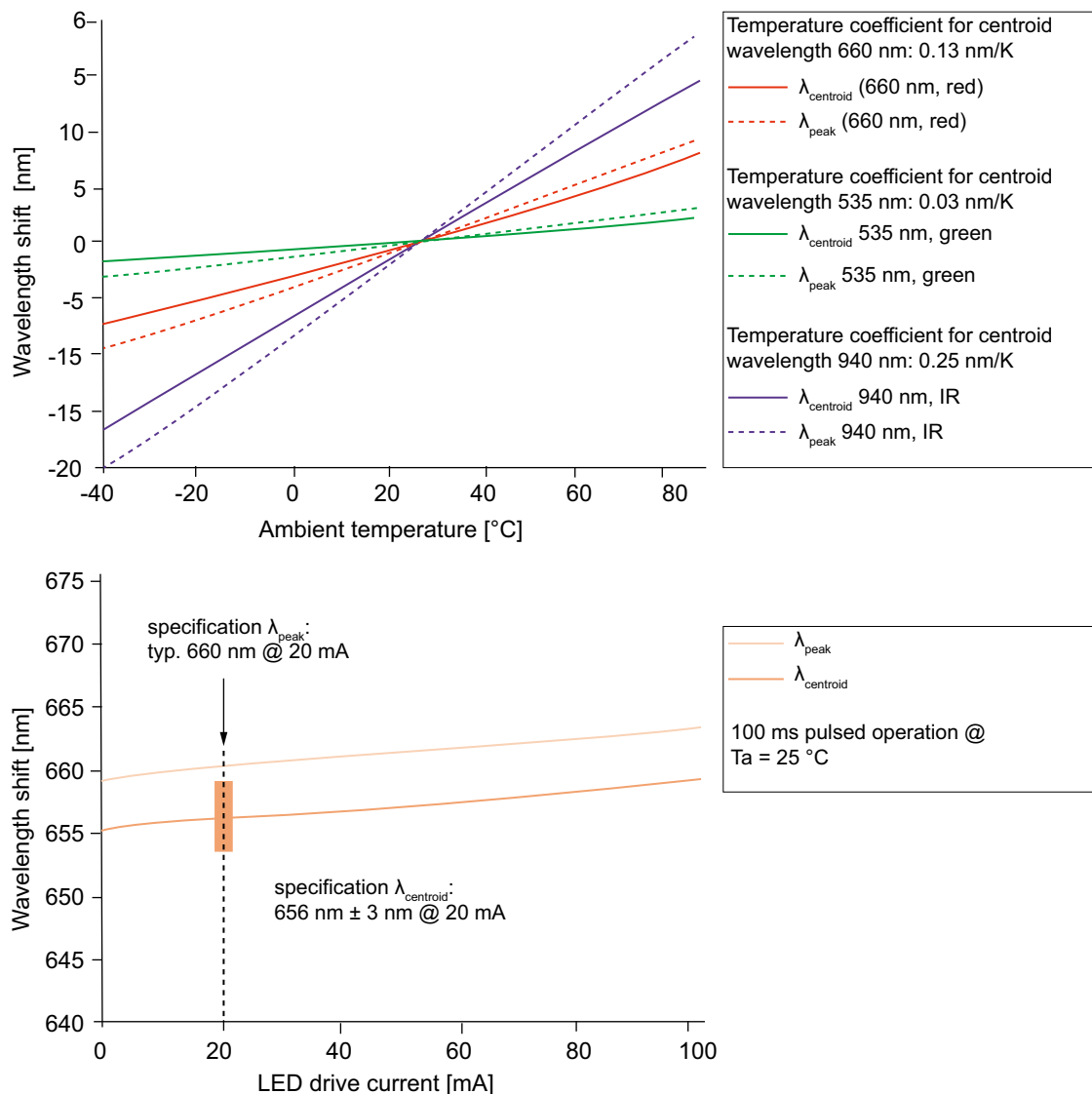
Figure 11: Asymmetric spectrum ($\lambda_{\text{peak}} \neq \lambda_{\text{centroid}}$) for defining the wavelength of an LED (example of the 940 nm IR LED)



5.4 The influence of wavelength shift

LEDs should have very tight wavelength specifications and no secondary peak. This ensures reproducible signal readings as the slope of the blood absorption coefficient (d_a/d_λ) is the highest. The LEDs should also feature a low temperature shift as well as a narrow spectral bandwidth. In general, the emitting wavelength depends on the ambient temperature and the driving conditions (pulse peak current, pulse width and duty cycle). Figure 12 shows an example of this temperature and driving current dependency. The temperature dependence of the emitter wavelength is relative to $T_a = 25^\circ\text{C}$ at 20 ms pulse. The wavelength relevant for the application is $\lambda_{\text{centroid}}$.

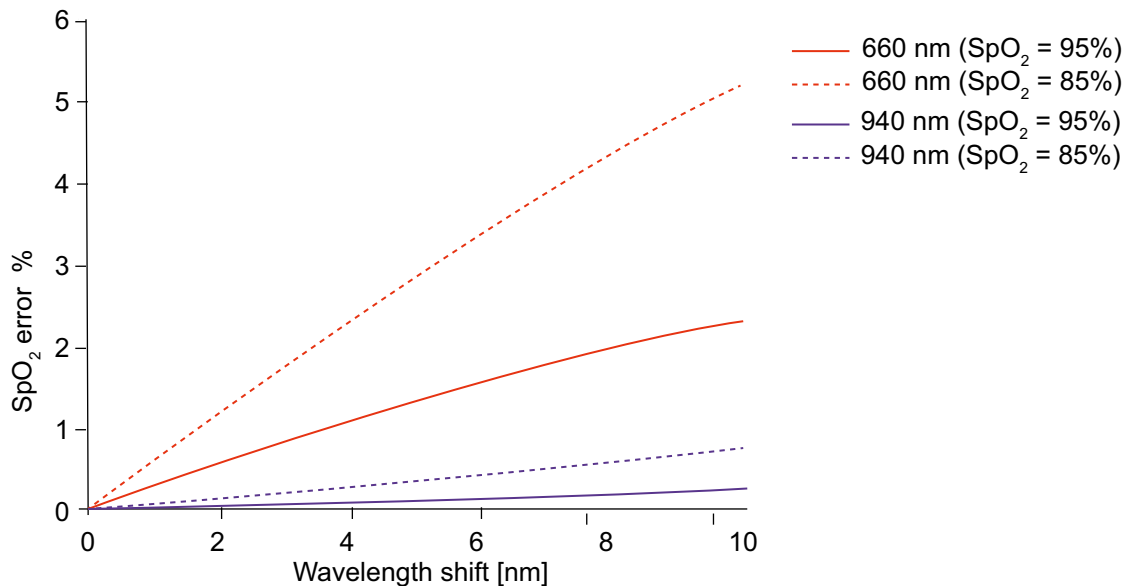
Figure 12: Temperature dependence of the emitter wavelength and drive current dependency



Using short pulses also minimizes any temperature-dependent wavelength shift as well as spectral broadening due to internal heating of the LED (e.g., pulse width $< 300\ \mu\text{s}$ and repetition rate $> 2\ \text{ms}$).

As typical for visible LEDs, the wavelength shift of the red LED is most critical and must be compensated. Figure 13 presents an example of the influence of the emitter wavelength shift vs. SpO₂ measurement accuracy. The lower the SpO₂ level, the higher the overall error due to the wavelength shift. In general, the wavelength stability of, for example, a 940 nm IR LED is non-critical compared to the 660 nm LED.

Figure 13: Example of a SpO₂ error due to wavelength shift of the emitter



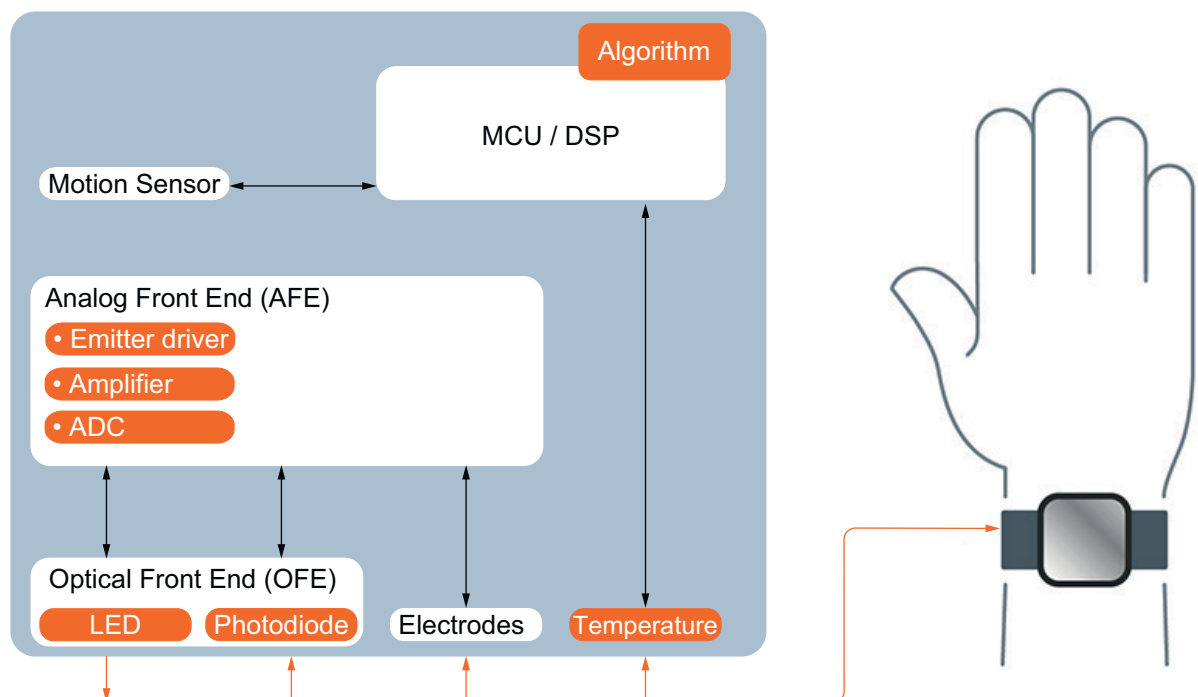
For highly accurate measurements it is recommended to compensate the wavelength shift of the LED. This can be done by e.g., monitoring the junction temperature ($\Delta\lambda = f(T_j)$) of LEDs. Implementation can be realized via ambient temperature measurement, either with a temperature sensor or e.g. via the junction voltage of an external Si-diode located close to the LED. This voltage is correlated to the ambient temperature, indicative for the wavelength shift of the LED during the particular operating conditions (calibration must be done e.g. during final device testing to obtain room temperature reference). Another, more complex method is via direct junction voltage measurement immediately after the LED pulse (e.g., biasing the LED with 1 μ A and measuring the forward voltage drop, similar to Si-diodes).

5.5 Operating conditions

The entire health monitoring system consists of various functional building blocks, as shown in Figure 14. ams OSRAM provides all the key components:

- Emitters
- Photodiodes
- Optical packages including LEDs and PDs
- Analog FrontEnd (AFE) (for example AS7056 & AS7057)
- Integrated modules
- Algorithms (selectively)
- Temperature sensors

Figure 14: Functional building blocks for a health monitoring system



The opto-electronic part in the building blocks describe the Optical Front End (OFE). It can be realized through discrete components (LEDs and photodiodes) or with an integrated module. The Analog Front End (AFE) provides the analog signal processing (photodiode signal amplification and analog-to-digital conversion) and programmable LED driving. Several commercial chipsets are available. An integrated solution can also be selected here, which provides LEDs, photodiodes and IC in one component.

Furthermore, the health monitoring system requires a suitable micro controller and a heart rate and motion compensation algorithm. In dynamic situations, motion sensors measure artefacts that arise from the user's motion. A motion compensation feature is therefore necessary to obtain an accurate PPG measurement under these circumstances.

There are (slightly) different measurement requirements concerning the application scenario:

- Heart rate only
- Heart rate with pulse oximetry

In case of heart rate only applications the DC component of the photocurrent can be neglected and only the periodicity of the AC component ($I_{\max} - I_{\min}$, frequency) is of interest. For pulse oximetry both the DC and the AC components ($I_{\min}$, $I_{\max}$) are required. Thus in general, a pure heart rate device is easier to implement as it requires only one LED. For most body locations the green LED might be the preferred choice. However, there is the option to drive the red as well as the IR LED. The IR LED might be advantageous as its light is invisible to the human eye. This can be a key criterion as in dark environments the green or red glow — if not shielded properly — might distract the user. In addition, the IR LED features the lowest forward voltage at comparable current density.

The signal level (signal quality) is affected by the measurement system as well as by biological characteristics.

The complete system includes the optical engine and sensor with LEDs for illumination and a photodiode for signal detection. The photocurrent can be split into a DC component (no information if only the heart rate is of interest) and the AC component. In addition, ambient light might be present (considered as AC+DC noise). Especially IR light can penetrate deeply into / through the skin. For example IR light from pulsed light sources (fluorescent or incandescent) or light from DC sources (like the sun) modulated by body movements can contribute to the optical signal as noise. The DC component of optical noise is usually subtracted due to an ambient light measurement immediately prior to or after the LED light measurement, resulting in an effective signal of:

$$I = I_{DC(signal)} + I_{AC(signal)} + I_{AC(noise)}$$

Important operating parameters influencing the signal quality:

- LED current
- LED on-time
- LED repetition rate

Increasing the LED current results in the following:

- $AC_{(pk-pk)} (= I_{max} - I_{min})$ increases
- DC ($= I_{min}$) increases
- AC/DC ratio stays the same
- $AC_{(pk-pk)}$ to ambient light ratio increases
- SNR increases ($AC_{(pk-pk)}$ to electronic noise ratio or dark current increases)
- The energy consumption increases

Reducing the on-time of the LED provides the following results:

- Comparable AC/DC ratio
- Comparable AC to ambient light ratio
- Comparable AC to dark current ratio
- Possible reduction of SNR due to (potentially) more electronic noise from a shorter integration time
- Higher required bandwidth for the transimpedance amplifier (TIA)
- The larger system bandwidth makes it difficult to filter out (ambient) noise (in case ambient subtraction is performed in the digital domain)
- Reductions in energy consumption

Using an even larger photodiode would not result in an improved AC/DC ratio.

Increasing the repetition rate of the LED might result in higher accuracy of the signal at the expense of higher energy consumption. In essence the sampling rate needs to be high enough not to miss the peaks / troughs of the pulsatile signal.

In order to increase the battery life of wearable or mobile devices the LED on-time and LED current (depending on the noise level and the signal amplification) should be reduced as much as possible.

Finally, chipsets are available which include the complete analog signal processing (incl. LED driver) as well as analog-to-digital conversion. Detectors can be directly connected to these chipsets to enable fast and easy evaluation and design.

5.6 Crosstalk reduction

Typical optical front-end (OFE) of the VSM-application consists of a light emitter (for example SFH 7018) and a photodetector (for example SFH 2202). Light from the emitter is coupled into the skin and spreads through the human tissue. It hits the blood vessels and gets modulated according to momentary amount of the blood in the vessels. This light is then detected by the

photodiode and the modulation is extracted from the signal by data processing algorithms. The challenging part of the whole solution is that modulation depth in the real application is very weak and there is high amount of light, seen by detector, but not modulated by the blood volume. This light spreads across the surface of the skin and does not penetrate deep into the tissue.

Additionally, the cross-talks between emitter and photodiode increase the amount of non-modulated light and, therefore, reduces the accuracy of the measurements of heart rate and blood oxygen saturation.

The SFH 7018 offers a solution of these problems by utilizing unique features:

- Two-cavity design with separated cavities for green and red/IR emitters (Figure 15). Each light source has a well-defined size (size of corresponding cavity), green for heart rate measurement (HRM) and red/infrared for blood oxygen saturation. An optimum distance between light emitter and photodiode, resulting in the highest useful signal, can be easily achieved by customer design.
- No side emission out of the package. All the light is emitted from the top surface of the package and can be used by the customer system for precise measurements. The system cross-talk is significantly reduced.
- Improved optical design of the package ensures highest efficiency in the application and allows reliable measurements of vital signs with significantly lower energy consumption.

Figure 15: SFH 7018: Two-cavity design with no side emission



6 Summary

A combination of LEDs and detectors provide the technical opto-electrical requirements for heart rate monitoring and measuring oxygen saturation. The light from the transmitter is coupled into the skin and is modulated according to the momentary amount of blood in the blood vessels. This signal is then detected by the photodiode. The complete health monitoring system consists of the Optical Front End (OFE) with LEDs and photodiodes, the Analog Front End (AFE) with analog signal processing and programmable LED driving, a suitable microcontroller and an algorithm for compensation of health monitoring functions and motion.

The quality of the measurement data is essential for the applications. Their accuracy and stability can be improved by:

- Placement close to the skin
- Measurement point with good blood circulation
- Compensation of LED wavelength shift
- Reduction of crosstalk between emitter and photodiode

ams OSRAM offers a comprehensive product portfolio to support individual health monitoring applications. For further questions, get in contact with our [technical customer service](#).

7 Glossary and abbreviations

Table 9 shows the abbreviations for this documentation.

Table 9: Abbreviations

Term	Description
AFE	Analog front-end
AGE	Advanced Glycation End Products
BioZ	Bioimpedance
ECG	Electrocardiogram
GSR	Galvanic Skin Response
HbO ₂	Oxygenated hemoglobin
HRM	Heart Rate Monitoring
IR	Infrared Light with wavelengths above 700 nm
LED	Light Emitting Diode
OFE	Optical Front End
PD	Photodiode
PPG	Photoplethysmography
PS	Proximity Sensing
RMSE	Root Mean Square Error
RRp	Respiratory Rate
SNR	Signal-to-Noise Ratio
SpO ₂	Peripheral Oxygen Saturation
VCSEL	Vertical Cavity Surface Emitting Laser
VSM	Vital sign monitoring

Table 10 shows the glossary for this documentation

Table 10: Glossary

Term	Description
Wearable	A portable electronic device for continuous monitoring of vital parameters
BIOFY™	A product line from ams OSRAM offering integrated optical modules for measuring vital parameters
Pulse Oximetry	Measurement of blood oxygen saturation using optical sensors

ABOUT ams OSRAM Group (SIX: AMS)

The ams OSRAM Group (SIX: AMS) is a global leader in intelligent sensors and emitters. By adding intelligence to light and passion to innovation, we enrich people's lives. With over 110 years of combined history, our core is defined by imagination, deep engineering expertise and the ability to provide global industrial capacity in sensor and light technologies. Our around 20,000 employees worldwide focus on innovation across sensing, illumination and visualization to make journeys safer, medical diagnosis more accurate and daily moments in communication a richer experience. Headquartered in Premstaetten/Graz (Austria) with a co-headquarters in Munich (Germany), the group achieved EUR 3.6 billion revenues in 2023. Find out more about us on <https://ams-osram.com>

DISCLAIMER

PLEASE CAREFULLY READ THE BELOW TERMS AND CONDITIONS BEFORE USING THE INFORMATION SHOWN HEREIN. IF YOU DO NOT AGREE WITH ANY OF THESE TERMS AND CONDITIONS, DO NOT USE THE INFORMATION.

The information provided in this general information document was formulated using the utmost care; however, it is provided by ams-OSRAM AG or its Affiliates* on an "as is" basis. Thus, ams-OSRAM AG or its Affiliates* does not expressly or implicitly assume any warranty or liability whatsoever in relation to this information, including – but not limited to – warranties for correctness, completeness, marketability, fitness for any specific purpose, title, or non-infringement of rights. In no event shall ams-OSRAM AG or its Affiliates* be liable – regardless of the legal theory – for any direct, indirect, special, incidental, exemplary, consequential, or punitive damages arising from the use of this information. This limitation shall apply even if ams-OSRAM AG or its Affiliates* has been advised of possible damages. As some jurisdictions do not allow the exclusion of certain warranties or limitations of liabilities, the above limitations and exclusions might not apply. In such cases, the liability of ams-OSRAM AG or its Affiliates* is limited to the greatest extent permitted in law.

ams-OSRAM AG or its Affiliates* may change the provided information at any time without giving notice to users and is not obliged to provide any maintenance or support related to the provided information. The provided information is based on special conditions, which means that the possibility of changes cannot be precluded.

Any rights not expressly granted herein are reserved. Other than the right to use the information provided in this document, no other rights are granted nor shall any obligations requiring the granting of further rights be inferred. Any and all rights and licenses regarding patents and patent applications are expressly excluded.

It is prohibited to reproduce, transfer, distribute, or store all or part of the content of this document in any form without the prior written permission of ams-OSRAM AG or its Affiliates* unless required to do so in accordance with applicable law.

* ("Affiliate" means any existing or future entity: (i) directly or indirectly controlling a Party; (ii) under the same direct, indirect or joint ownership or control as a Party; or (iii) directly, indirectly or jointly owned or controlled by a Party. As used herein, the term "control" (including any variations thereof) means the power or authority, directly or indirectly, to direct or cause the direction of the management and policies of such Party or entity, whether through ownership of voting securities or other interests, by contract or otherwise.)



For further information on our products please see the Product Selector and scan this QR Code.

Published by ams-OSRAM AG
Tobelbader Strasse 30, 8141 Premstaetten, Austria
ams-osram.com © All Rights Reserved.

Published by ams-OSRAM AG

Tobelbader Strasse 30,
8141 Premstaetten Austria

Phone +43 3136 500-0

ams-osram.com

© All rights reserved

